

6 Proof by Contradiction

Use the method of proof by contradiction to prove the following statements. (In each case, you should also think about how a direct or contrapositive proof would work. You will find in most cases that proof by contradiction is easier.)

2. Suppose $n \in \mathbb{Z}$. If n^2 is odd, then n is odd.

Proof. For the sake of contradiction, suppose n^2 is odd and n is not odd. Then n^2 is odd and n is even. Since n is even, $n = 2c$ for some $c \in \mathbb{Z}$. Then $n^2 = (2c)^2 = 4c^2 = 2(2c^2)$, so n^2 is even. Thus n^2 is odd and n^2 is even, a contradiction. $\square$

4. Prove that $\sqrt{6}$ is irrational.

Proof. For the sake of contradiction, suppose $\sqrt{6}$ is not irrational. Then $\sqrt{6}$ is rational. Then $\sqrt{6} = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$, where $\frac{a}{b}$ is fully reduced; in particular, a and b are not both even. Squaring both sides gives $6 = \frac{a^2}{b^2}$, thus $a^2 = 6b^2 = 2(3b^2)$. Then a^2 is even, which implies a is even. Since a and b are not both even, it follows that b is odd, and hence b^2 is odd. Since a is even, $a = 2k$ for some $k \in \mathbb{Z}$. Substituting this value for a in the equation we get $(2k)^2 = 2(3b^2)$, so $4k^2 = 2(3b^2)$, and dividing both sides by 2 gives $2k^2 = 3b^2$. Thus $3b^2$ is even. But b^2 is odd and 3 is odd, and the product of two odd numbers is odd, so $3b^2$ is odd. Thus $3b^2$ is both even and odd, a contradiction. $\square$

6. If $a, b \in \mathbb{Z}$, then $a^2 - 4b - 2 \neq 0$.

Proof. For the sake of contradiction, suppose $a, b \in \mathbb{Z}$ but $a^2 - 4b - 2 = 0$. Then we have $a^2 = 4b + 2 = 2(2b + 1)$, which means a^2 is even. Therefore a is even, so $a = 2c$ for some $c \in \mathbb{Z}$. Substituting this value for a back in the equation we get

$$(2c)^2 - 4b - 2 = 0$$

$$4c^2 - 4b - 2 = 0$$

$$4(c^2 - b) = 2$$

$$2(c^2 - b) = 1$$

Since $c^2 - b \in \mathbb{Z}$, the last equation says that 1 is an even number, a contradiction. $\square$

8. Suppose $a, b, c \in \mathbb{Z}$. If $a^2 + b^2 = c^2$, then a or b is even.

Proof. For the sake of contradiction, suppose that $a^2 + b^2 = c^2$, and that a and b are odd. Then there exist integers m and n for which $a = 2m + 1$ and $b = 2n + 1$. Hence, $a^2 = (2m + 1)^2 = 4(m^2 + m) + 1$, so $a^2 \equiv 1 \pmod{4}$. Similarly $b^2 \equiv 1 \pmod{4}$. Since $a^2 + b^2 = c^2$, it follows that $c^2 \equiv 1 + 1 \equiv 2 \pmod{4}$. But every square is congruent to either 0 or 1 $\pmod{4}$, so $c^2 \not\equiv 2 \pmod{4}$, a contradiction. $\square$

10. There exist no integers a and b for which $21a + 30b = 1$.

Proof. For the sake of contradiction, assume there exist integers a and b for which $21a + 30b = 1$. Then $1 = 3(7a + 10b)$. Since $7a + 10b \in \mathbb{Z}$, this implies $3 \mid 1$, a contradiction. $\square$